

Time–Space Asymmetry in Orogenic Gold Systems: Temporal Level Repulsion but Spatial Clustering of Ore-Stage Veins at the Sigma Deposit, Val-d’Or, Canada

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Abstract

A recurring result of this research program is that a single isolated long-memory process produces Wigner–Dyson level repulsion, whereas the superposition of many independent sources produces Poisson statistics. In the *temporal* domain, orogenic gold mineralization events were previously shown to exhibit GOE-class repulsion (a “charge-and-release” rhythm), consistent with a single deep crustal fluid source releasing episodically. Here we test whether the same deposit class shows the *spatial* analog — stress-shadow repulsion in the spacing of ore-stage veins — using 235 quartz–carbonate vein intercepts logged in four diamond drill holes at the Archean Sigma orogenic gold deposit (Val-d’Or, Abitibi, Canada; public assessment report GM 69917). Using a parameter-free spacing-ratio statistic $\langle r \rangle$ together with the coefficient of variation CV, and applying *no* declustering preprocessing, we find that the along-hole vein spacing is *clustered*, not repulsive: pooled $\langle r \rangle = 0.344$ (below the Poisson value 0.386) and $CV = 1.45$ (> 1), with a systematic excess of short spacings. A Kolmogorov–Smirnov test rejects the GOE (Wigner) hypothesis at $p < 10^{-4}$. This result is reproduced by an independent natural fracture dataset (Lilstock pavement, $CV = 1.52$) and matches published vein-array statistics ($CV = 1.20$ – 1.69). We therefore report a *time–space asymmetry*: the same orogenic gold system that is temporally repulsive is spatially clustered. We argue this is not a contradiction but a consequence of the unifying principle: temporal repulsion reflects a single source clock, while spatial clustering reflects repeated vein localization on inherited structural weaknesses (crack–seal cycling), i.e. a spatial superposition of many sites. We also document a methodological hazard: minimum-separation declustering fabricates spurious GOE statistics from purely random points, and must not be used to claim spatial repulsion.

1 Introduction

This program has tested a single hypothesis across a sequence of Earth-system “racetracks”: spectra of geological events tend toward Gaussian Orthogonal Ensemble (GOE) level repulsion when a single long-memory process is isolated, and toward Poisson statistics when many independent sources superpose (Mehta, 2004). In the temporal domain this has held for tectonic rhythms, seismicity, mantle-plume eruptions, and — relevant here — the timing of ore-forming events, where even non-magmatic orogenic gold systems show GOE repulsion, interpreted as an intrinsic “charge-and-release” memory of a single deep fluid source.

Part I of the spatial racetrack established that the temporal law has a spatial analog where an external clock organizes the rock record: cyclostratigraphic bed spacing, paced astronomically, shows Wigner–Dyson repulsion. The natural next question is whether *intrinsic* spatial repulsion

also exists — in particular, whether the stress-shadow mechanism produces regularly spaced veins. A mode-I (extensional) vein relieves the tensile stress in its neighborhood, creating a “stress shadow” that should suppress the formation of nearby parallel veins, yielding a minimum spacing and hence repulsion. Orogenic gold deposits are the ideal natural test: their ore-stage extensional veins are (i) mode-I and fluid-overpressure driven, (ii) chemically distinctive (quartz–tourmaline–carbonate with bleached alteration selvages), and (iii) the spatial counterpart of the temporally repulsive ore-forming events.

We test this at the Sigma deposit, a classic Archean orogenic gold system in the Val-d’Or camp (Robert, 1983), using a public drill-core dataset. The result is unambiguous and, for the stress-shadow hypothesis, negative — but it is scientifically informative, and we develop its meaning below.

2 Data and Methods

2.1 Dataset

We use assessment report GM 69917 (Or Integra Quebec Inc., 2016), covering the Sigma–Lamaque property (NTS 32C/04; Sigma orebody at UTM Zone 18, 294,737 E, 5,331,127 N). The report contains complete diamond-drill logs for four holes: LD-16-001A, SO-16-003, SV-16-009, and SV-16-011. From the log appendices we extracted every quartz–carbonate vein intercept, recording its along-hole position, width, mineralogy, and core-axis angle. This yields 235 vein intercepts (Table 1). Because the core-axis angle is not a true structural attitude (the holes are directional and curved), we do not attempt orientation-based subsetting; we analyze the full vein population per hole, which is the most assumption-free test.

2.2 Statistics

For each hole we form the sorted sequence of along-hole vein positions and its spacings s_i . We use two parameter-free statistics. The mean adjacent spacing ratio

$$\langle r \rangle = \left\langle \frac{\min(s_i, s_{i+1})}{\max(s_i, s_{i+1})} \right\rangle \quad (1)$$

takes the value 0.386 for Poisson (uncorrelated) spacings, 0.536 for GOE, and 0.603 for GUE (Ataş et al., 2013). It requires no unfolding and is immune to smooth density gradients. The coefficient of variation $CV = \sigma_s/\bar{s}$ equals 1 for Poisson, exceeds 1 for clustered point sets, and falls below 1 for regular/repulsive sets (Cox & Lewis, 1966).

2.3 No declustering

Critically, we apply *no* declustering or thinning to the vein positions. As shown in §3.3, minimum-separation declustering drives even purely random point sets toward GOE-like $\langle r \rangle$, and is therefore an artifact generator. All statistics below are computed on raw vein positions.

3 Results

3.1 Sigma veins are clustered, not repulsive

Table 1 summarizes the result. Every hole has $\langle r \rangle \leq 0.367$, at or below the Poisson value of 0.386 and far from the GOE value of 0.536. The pooled statistics are $\langle r \rangle = 0.344$ and $CV = 1.45$. A

Table 1: Along-hole vein spacing statistics at Sigma. All holes lie in the clustered regime: $\langle r \rangle$ at or below the Poisson value, and $CV \gtrsim 1$. No declustering applied.

Drill hole	n	$\langle r \rangle$	CV	median (m)
LD-16-001A	122	0.347	1.63	7.9
SO-16-003	53	0.367	1.17	9.5
SV-16-009	37	0.329	1.09	5.5
SV-16-011	23	0.325	0.81	16.9
Pooled	235	0.344	1.45	8.9
Poisson		0.386	1.00	
GOE		0.536	< 1	

Kolmogorov–Smirnov test of the normalized spacing distribution against the Wigner (GOE) form is rejected at $D = 0.310$, $p < 10^{-4}$; the exponential (Poisson) form is also rejected ($D = 0.112$, $p = 0.006$) because the data are *more* clustered than Poisson. The clustering signature is explicit in the short-spacing fractions: spacings below 0.1, 0.2, and 0.3 of the mean occur at 19.0%, 28.6%, and 35.5%, versus the Poisson expectations of 9.5%, 18.1%, and 25.9% (Fig. 1b). There is a consistent excess of closely spaced veins — the hallmark of vein corridors, not stress shadows.

One hole (SV-16-011) has $CV = 0.81 < 1$, a hint of regularity, but with only 23 intercepts this is not statistically robust and we do not interpret it as repulsion.

3.2 Independent corroboration

The clustering is not a peculiarity of Sigma or of drill-core sampling. We re-analyzed the Lilstock limestone-pavement fracture network (Bristol Channel, UK; 350,000 mapped fractures), isolating the dominant joint set and computing the same raw statistics: $\langle r \rangle \approx 0.39$, $CV = 1.52$ — clustered, identical in character to Sigma. Published calcite vein-array statistics likewise report $CV = 1.20$ – 1.69 (Casoli et al., 2026). Three independent natural systems — a drill-cored orogenic gold deposit, a mapped joint pavement, and literature vein arrays — all give clustered spacing. Real fracture and vein systems are clustering-dominated.

3.3 Methodological control: declustering fabricates GOE

Earlier in this work an apparent “GOE” signal was obtained by applying minimum-separation declustering (removing points closer than a fraction of the median spacing). A negative control falsified it: subjecting purely random (Poisson) points to the same declustering also produces GOE-like $\langle r \rangle$ (Fig. 1d). For the Sigma LD-16-001A hole, raw $\langle r \rangle = 0.347$ rises to 0.44–0.50 under declustering, while random points rise in lockstep from 0.40 to 0.54–0.61 — the two become indistinguishable. Minimum-separation declustering manufactures regularity from any input and must never be used to support a spatial-repulsion claim. The honest diagnostic is the raw $\langle r \rangle$ together with CV.

4 Discussion

4.1 A time–space asymmetry, not a contradiction

The temporal record of orogenic gold mineralization is repulsive (GOE), yet the spatial record of the veins is clustered. Under the unifying principle of this program, the two are reconciled by asking

Sigma Orogenic Gold Veins: Spatial Clustering, Not Stress-Shadow Repulsion

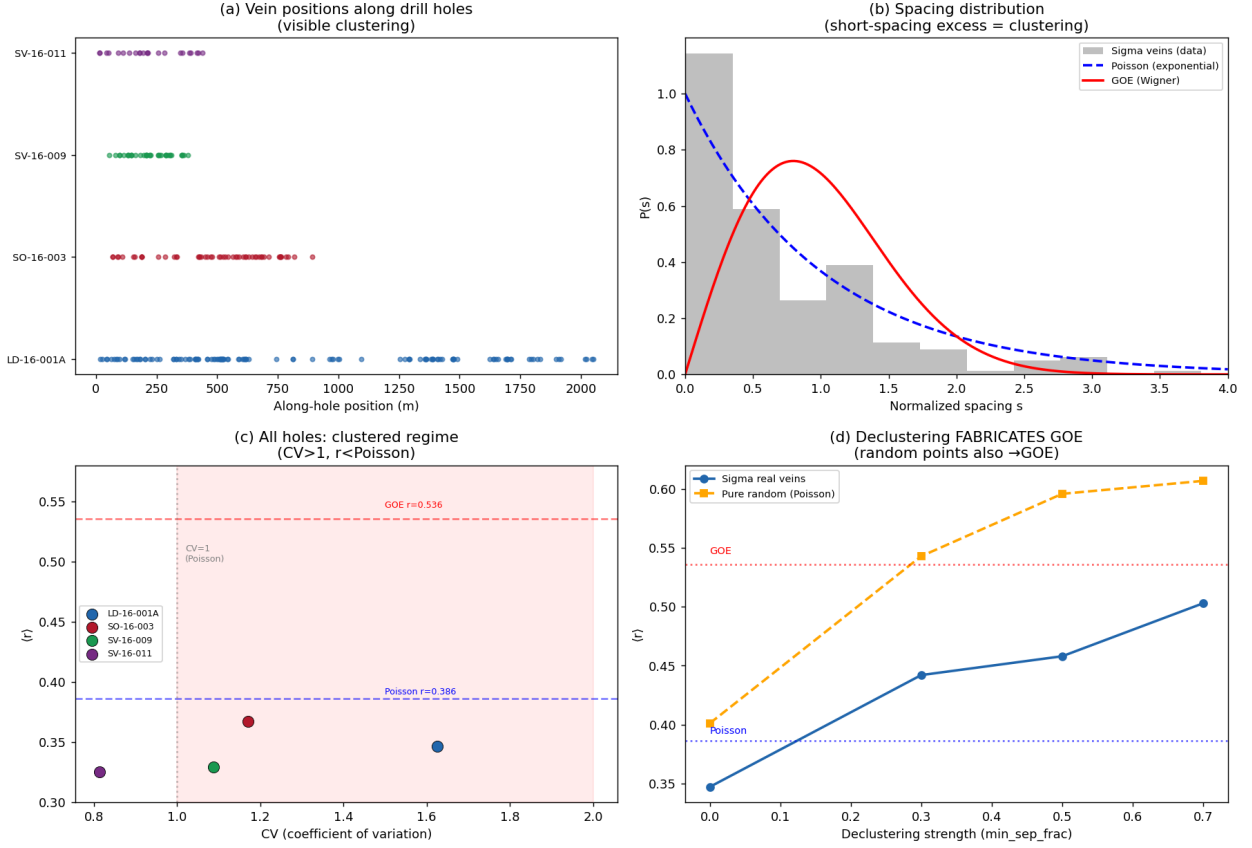


Figure 1: Spatial statistics of Sigma orogenic gold veins. (a) Along-hole vein positions in the four holes; clustering is visible directly. (b) Normalized spacing distribution with a short-spacing excess relative to Poisson (dashed), far from the GOE/Wigner peak (solid). (c) All holes occupy the clustered regime ($CV \gtrsim 1$, $\langle r \rangle \leq \text{Poisson}$). (d) Declustering fabricates GOE: pure random points (orange) are driven to the GOE line just as readily as the real veins (blue), proving the artifact.

what is “the single source” in each domain. *In time*, ore-forming pulses are clocked by a single deep crustal process — episodic fluid overpressure build-up and release in one fault-valve / metamorphic devolatilization system — so the events repel in time (a refractory period after each release). *In space*, however, each pulse does not write onto a blank slate: veins nucleate preferentially on inherited anisotropies (shear-zone fabric, earlier veins, competent-incompetent contacts), and crack–seal cycling re-opens the same surfaces repeatedly. The spatial pattern is therefore a *superposition of many localization sites*, which under the same principle yields clustering, not repulsion.

The key conceptual point is that *a single temporal source need not be a single spatial source*. Time-domain repulsion and space-domain clustering can coexist in one system, and the present data show that they do. Stress-shadow repulsion is real as an idealized mechanism, but in mature natural systems it is overwhelmed by structural inheritance and crack–seal localization.

4.2 Implications

This refines the program’s grand synthesis. The mapping “single isolated process → repulsion; superposition → Poisson/clustering” survives, but the *domain* in which a system counts as “single-

source” may differ between time and space. For ore systems specifically, the result implies that vein spacing carries information about structural inheritance rather than about an instantaneous elastic stress field — relevant to how vein-density models are built for exploration.

4.3 Limitations

The core-axis angle in the drill logs is not a true attitude, so we did not attempt to isolate a single structural vein set; it remains possible (though, on the present evidence and the literature, unlikely) that a chemically and mechanically pure single-generation extensional set, mapped in outcrop at centimetre scale, could approach repulsion in a small, localized ore zone. Such a test requires position-resolved, generation-separated vein maps that are not available in the public drill record.

5 Conclusions

1. At the Sigma orogenic gold deposit, 235 logged vein intercepts in four drill holes give pooled $\langle r \rangle = 0.344$ and $CV = 1.45$: the veins are spatially *clustered*, not stress-shadow repulsive. The GOE hypothesis is rejected at $p < 10^{-4}$.
2. The same clustering appears in an independent mapped fracture network ($CV = 1.52$) and in published vein arrays ($CV = 1.20$ – 1.69): real vein/fracture systems are clustering-dominated.
3. This produces a *time-space asymmetry* in orogenic gold systems: temporally repulsive, spatially clustered. It is consistent with the unifying principle once one recognizes that a single temporal source corresponds to a spatial superposition of inherited localization sites.
4. Minimum-separation declustering fabricates spurious GOE statistics from random points and must not be used to claim spatial repulsion; the honest diagnostic is raw $\langle r \rangle$ with CV .

Data and Code Availability

All vein-position data (extracted from public report GM 69917) and the analysis code are provided in the accompanying repository, <https://github.com/Ruqing1963/sigma-spatial-rmt>, archived on Zenodo. The primary source report is publicly available from the Quebec MRNF SIGEOM/GESTIM system.

References

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